

A Report on

**EFFICIENT COMPUTER SPECIFICATIONS RETRIEVAL USING
RFID AND STM32**

for

Mini Project 2A (REV- 2019 'C' Scheme) of Third Year, (TE Sem – V)

in

Electronics & Telecommunication Engineering

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UNIVERSITY OF MUMBAI

AY 2023-2024



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CERTIFICATE

This is to certify that the project entitled **EFFICIENT COMPUTER SPECTIFICATION RETRIVAL USING RFID AND STM32** is a bonafide work of

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submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Mini Project 2A (REV- 2019 'C' Scheme) of Third Year, (TE Semester – V) in Electronics & Telecommunication Engineering** as laid down by **University of Mumbai** during academic year **2023-24**

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ABBREVIATION

The following tables describe the significance of various abbreviations used in the report.

Abbreviation	Meaning
RFID	Radio Frequency Identification
LCD	Liquid-crystal display

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1. INTRODUCTION

1.1 NECESSITY

Efficient computer specification retrieval has become essential for both individuals and organizations in the continually changing world of technology. Efficient computer specification retrieval has become essential for both individuals and organizations in the continually changing world of technology. There is now a wide range of solutions on the market due to the quick improvements in computer technology and components. However, having so many options also makes it difficult to determine which computer specs are best for a given set of requirements. For this reason, having a fast and effective procedure for obtaining precise and current computer specs is crucial.

In many situations, retrieving computer specifications efficiently is essential. For people, it guarantees that the parts they choose for their computer systems are appropriate and fit their needs. For professionals, designers, and gamers who depend on high-performance devices, this is extremely important. In the same way, for companies, effective

1.2 DEFINITION

Efficient computer specification retrieval refers to the ability to quickly and accurately retrieve detailed information about computer components such as processors, memory, storage, etc. using RFID technology and the STM32 microprocessor. With the use of RFID (Radio Frequency Identification) technology, a tag affixed to a computer component and a reader may communicate wirelessly. The STM32 microprocessor, with its advanced processing capabilities, serves as the brain of the system, facilitating efficient data processing and retrieval. By leveraging RFID and the STM32 microprocessor, computer specification retrieval can be automated and streamlined, saving time and improving accuracy

2. COMPARATIVE STUDY

RFID technology is the most efficient tool used for inventory tracking. Unlike the barcode scanning method, RFID technology provides a non-line-of-sight tracking solution that can be more reliable, faster, and more secure. Therefore RFID technology provides a faster and more efficient way of retrieving computer system specifications.

3. PROBLEM STATEMENT

There are several obstacles that impede the efficiency and productivity of the existing manual approach for collecting computer specifications. Manually looking up and documenting specifications takes time and is frequently prone to human mistakes. Furthermore, it is challenging to effectively access and compare specification data in the absence of a centralized database. Delays in decision-making result from this, as do possible compatibility problems and higher expenses from choosing the wrong or insufficient components.

An effective computer specification retrieval system utilizing STM31 and RFID is required in order to overcome these obstacles and improve process efficiency as a whole. Organizations can cut down on the time and effort needed to obtain reliable information by centralizing the specification data and automating the retrieval process. This, in turn, leads to improved decision-making, enhanced productivity, and cost savings.

4. MINI PROJECT DESIGN

4.1 BLOCK DIAGRAM

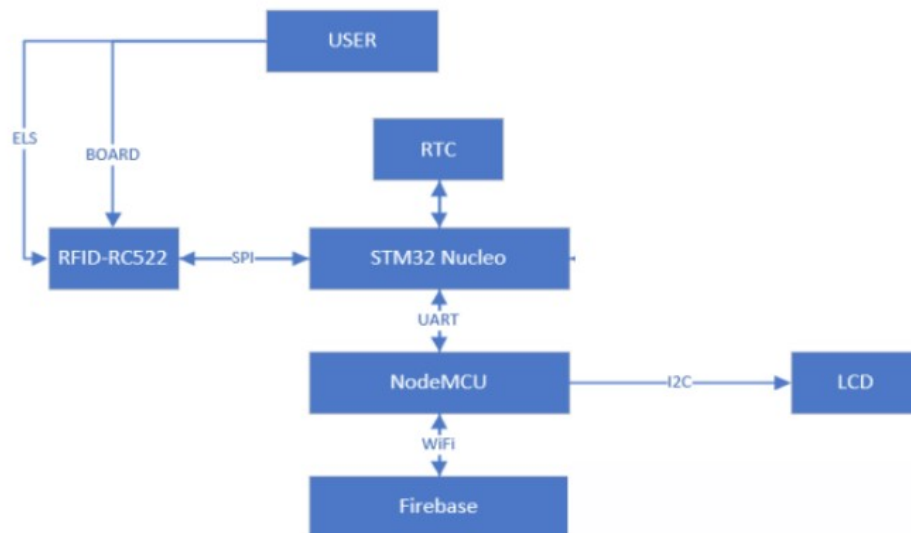


Fig 4.1.1 Block Diagram

4.2 BLOCK LOCK DESCRIPTION

The proposed system will consist of an RFID reader, an STM32 microcontroller board, and a user interface. The RFID reader will read the data from the RFID tags and send it to the user interface. The STM32 microcontroller board will process the data and store it in a database. The user interface will be used to view the recorded data.

4.3 WORKING

Certainly! Here's a concise summary of how the RFID interfacing project with STM32 and EM-18 reader module works:

1. Initialization

STM32 initializes UART communication and necessary peripherals.

The EM-18 module is powered up and ready to receive commands.

2. RFID Card Detection

When an RFID card or tag is brought near the EM-18 module, it emits radio waves.

3. RFID Data Transmission:

The EM-18 module detects the RFID card and reads its unique ID.

EM-18 sends the RFID card ID data to the STM32 microcontroller via UART communication.

4. UART Communication:

STM32 receives the RFID card ID data from the EM-18 module through UART.

UART protocol ensures the data is transmitted and received accurately.

5. Data Processing:

STM32 processes the received RFID data, validating its integrity.

The microcontroller compares the received RFID card ID with pre-defined IDs stored in its memory.

6. Access Control or Action

If the received RFID card ID matches a stored ID, the system grants access or performs a specific action.

Access control mechanisms, such as opening a door or allowing entry, are activated based on RFID card authentication.

7. Response Output:

The system provides a response, indicating whether access is granted or denied.

The response can be in the form of LEDs turning on/off, messages displayed on a screen, or any other desired output mechanism.

8. Repeat Detection:

The system continues to detect RFID cards as long as the EM-18 module is powered and active.

It processes each card's ID in the same manner, allowing or denying access based on the stored IDs.

9. End of Operation:

The system can be programmed to enter a low-power mode or shut down when not in use to conserve energy.

In summary, the project works by detecting RFID cards, transmitting their IDs to the STM32 microcontroller, processing the data, and triggering appropriate actions based on the authentication result.

5 COMPONENTS/TOOLS TO BE USED

5.1 COMPONENTS

1. STM32F103C8T6

The STM32F103C8T6 is a medium-density performance line, ARM Cortex-M3 32bit microcontroller in 48 pin LQFP package. It incorporates a high-performance RISC core with 72MHz operating frequency, high-speed embedded memories, an extensive range of enhanced I/Os and peripherals connected to two APB buses. The STM32F103C8T6 features 12bit ADC, timers, PWM timer, and standard and advanced communication interfaces.

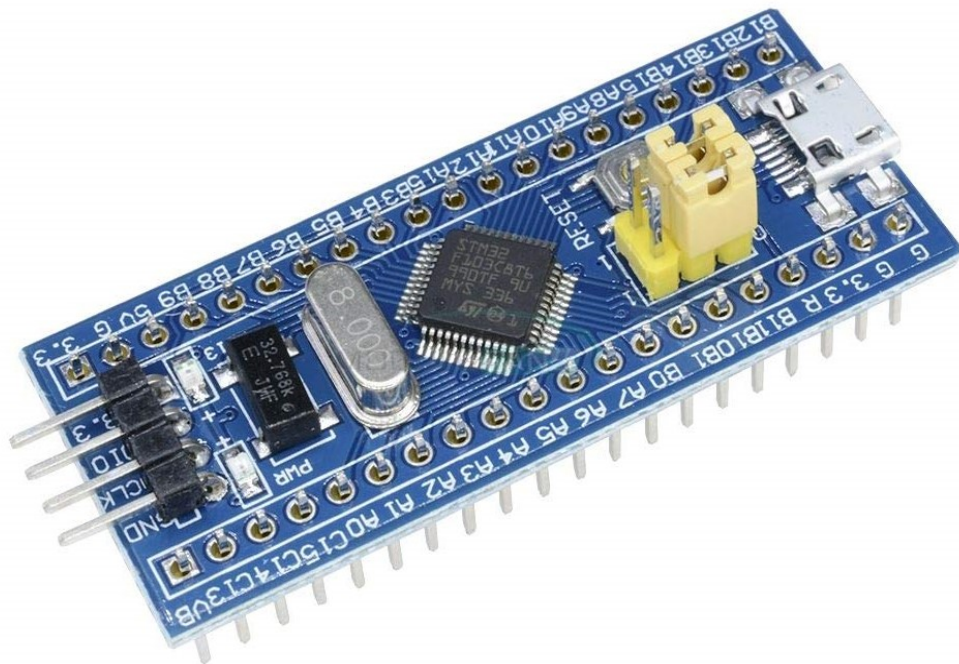


Fig 5.1.1 STM32F103C8T6 Microcontroller

2. RFID READER234

An RFID reader is a radio frequency device that emits a signal through an antenna. This signal is received by RFID tags that respond to interrogation by the reader. Responses are read by the reader, and through a variety of protocols the reader can communicate with all the RFID tags in its field.



Fig 5.1.2 RFID Reader

3. RFID Tags

An RFID tag in its most simplistic form, is comprised of two parts – an antenna for transmitting and receiving signals, and an RFID chip (or integrated circuit, IC) which stores the tag's ID and other information. RFID tags are affixed to items in order to use an RFID reader and antenna.



Fig 5.1.3 RFID Tags

4. NodeMCU

NodeMCU is a low-cost open-source IoT platform. It initially included firmware which runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which was based on the ESP-12 module. Later, support for the ESP32 32-bit MCU was added.



Fig 5.1.4 NodeMCU

5. 16 x 2 LCD Display

A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in a 5x7 pixel matrix. The 16 x 2 intelligent alphanumeric dot matrix display is capable of displaying 224 different characters and symbols.

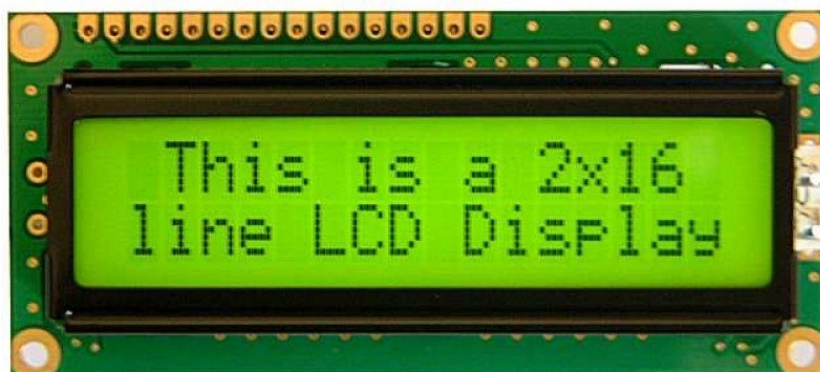


Fig 5.1.5 16 x 2 LCD Displa

6. Jumper Wires

A jumper wire is an electric wire that connects remote electric circuits used for printed circuit boards.

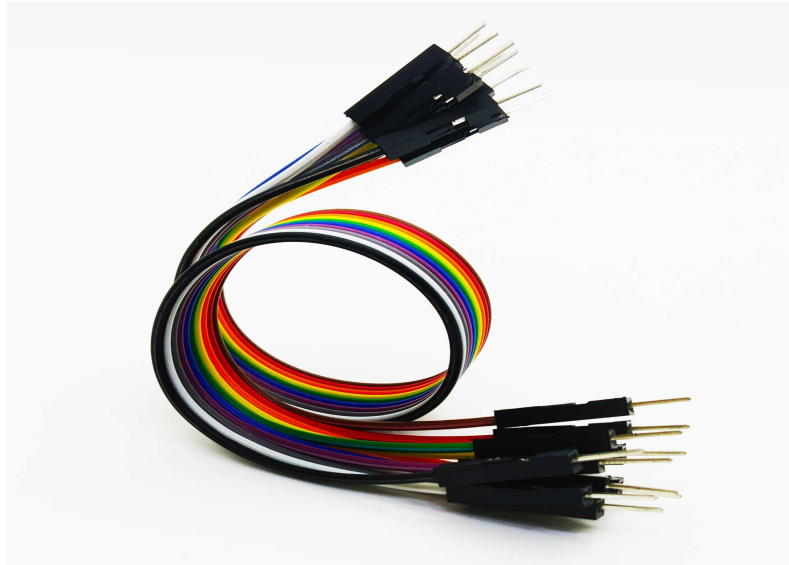


Fig 5.1.6 Jumper Wires

5.2 SOFTWARE

Arduino IDE



Fig 5.2.1 Arduino IDE Logo

6 PROPOSED EXECUTIONS STEPS

1. Conduct a thorough analysis of the existing computer specification retrieval process to identify pain points and areas for improvement.
2. Determine the key computer components that need to be tagged with RFID tags for effective tracking and identification.
3. Develop and implement an RFID tagging system, ensuring compatibility with different computer components and considering factors such as tag placement and durability.
4. Integrate STM31 technology into the system to capture, store, and manage computer specifications in a centralized database.
5. Establish a robust and scalable database infrastructure that allows for easy retrieval, comparison, and analysis of computer specifications.
6. Test and validate the efficiency, accuracy, and reliability of the system through pilot projects and user feedback.
7. Provide comprehensive training to users on the new system, including RFID tag management, STM31 device operation, and database utilization.
8. Develop documentation and guidelines for troubleshooting common issues, ensuring smooth operation and minimizing downtime.
9. Continuously monitor and evaluate the system's performance, making necessary adjustments and improvements as required.

7 TROUBLESHOOTING

7.1 PROBLEMS/FAULTS IN THE PROJECT

We run into a number of issues that might potentially prevent Efficient computer specifications retrieval using RFID and STM32 from operating as intended. Here are some of the issues:

1. LCD display not showing anything
2. RFID interfacing problem

7.2 STEPS TO SOLVE PROBLEMS/FAULTS IN THE PROJECT

Step 1: Information Gathering: Record all pertinent project information, including the hardware and software components that were utilized.

Step 2: Split and Take Over Separate out each problem and deal with it one at a time. Prioritize the LCD problem before addressing the RFID problem.

Step 3: Examine and assess

Examine each issue's code, connections, and component specs closely. Try using more straightforward code or fundamental features to identify the issue.

Step 4: Utilise debugging tools

To find the cause of issues, use debugging tools and methods, including oscilloscopes, LEDs, and print statements.

Step 5: Review the Records:

To be sure that your components and microcontroller are configured and used properly, check the datasheets, reference manuals, and documentation.

Step 6: Seek Online Resources:

Search online forums, communities, and documentation for similar issues and solutions

related to your project.

Step 7: Seek Expert Help:

If you're unable to resolve the issues, consider reaching out to experts or colleagues who may have experience with similar projects.

Step 8: Iterative Testing:

Make small changes and test them incrementally to identify the exact point of failure.

Step 9: Document Progress:

Keep a detailed record of your progress, changes made, and results at each step to help troubleshoot and prevent similar issues in the future.

Step 10: Implement Solutions:

Once you identify the root causes of the issues, implement the necessary changes to your code, connections, or hardware setup to resolve them.

CONCLUSION

In conclusion, the implementation of an efficient computer specification retrieval system using RFID technology and the STM32 microcontroller offers a promising solution to streamline the process of obtaining accurate and up-to-date computer specifications. This project addresses the need for a more efficient and error-free method, benefiting individuals, professionals, and organizations. The integration of RFID technology enhances speed, accuracy, and security, surpassing traditional methods like barcode scanning. Leveraging the STM32 microcontroller effectively processes and manages data, making it easily accessible. The system can overcome manual data retrieval limitations, reduce errors, eliminate compatibility issues, and lead to better decision-making and cost savings. Additionally, it is scalable and adaptable for various applications, extending its usefulness across different domains. Continuous monitoring and user training will be essential for optimal performance, making this system a significant advancement in accurate and efficient data retrieval in the technology world.

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